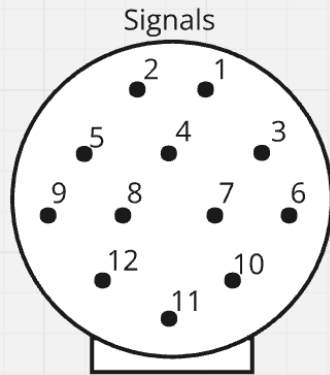


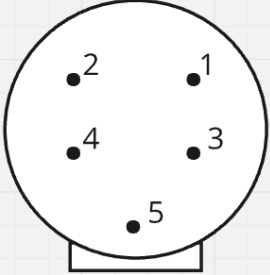
12-Prong Signal Connector

This sheet is defining the gray 12-pin signal cable going to the tool-head. It is meant for carrying **low amperage voltages only**. The pins are defined 1-12 and can be seen in the diagram below and also on the end of the connectors themselves.

Connector Pin #	Wire Color	Toolhead Connection	Controller Connection	Function (Thermoplastics)	Function (Ceramics)
1	Yellow	Ext. Nozzle/K-probe 2 (+)	Tmp. Controller 1 (pin)	Positive terminal for extended nozzle's temperature sensor	-
2	Purple	-	-	-	-
3	Dark Brown	Chamber/K-probe 1 (-)	Tmp. Controller 1 (pin)	Negative terminal for heating chamber's temperature sensor	-
4	Grey	-	-	-	-
5	Pink	-	-	-	-
6	Blue	-	-	-	-
7	White	(CL57T) DIR+	ClearCore I/O-1 (+)	-	Positive terminal for motor direction
8	Black	(CL57T) PUL-	ClearCore I/O-0 (signal pin*)	-	Signal terminal for motor stepping
9	Light Brown	Ext. Nozzle/K-probe 2 (-)	Tmp. Controller 2 (pin)	Negative terminal for extended nozzle's temperature sensor	-
10	Orange	Chamber/K-probe 1 (+)	Tmp. Controller 2 (pin)	Positive terminal for heating chamber's temperature sensor	-
11	Green	(CL57T) DIR-	ClearCore I/O-1 (signal pin*)	-	Signal terminal for motor direction
12	Red	(CL57T) PUL+	ClearCore I/O-0 (+)	-	Positive terminal for motor stepping

*"signal" is the blank terminal on ClearCore



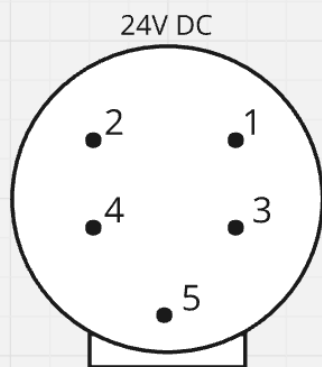
5-Prong 120V-AC Connector				
This Cable is Dangerous				
This sheet is defining the black 5-pin cable carrying 120V of alternating current (AC) going to the tool-head. DO NOT CONFUSE IT FOR THE 24V 5-PIN LABELED WITH BLUE TAPE. The system may short-circuit if you plug in 120VAC into 24VDC equipment.				
The pins are defined 1-5 and can be seen in the diagram below and also on the end of the connectors themselves. The "connection" refers to the end of the cable that is wired to a device in the robot controller.				
Connector Pin #	Toolhead Connection	Controller Connection	Function (Thermoplastics)	Notes
1	Chamber Hot Lead 1	120AC Neutral	Complete's circuit for SSR 1	Directly plugged into 120AC neutral
2	Ext. Nozzle Hot Lead 1	120AC Neutral	Complete's circuit for SSR 2	Directly plugged into 120AC neutral
3	Chamber Hot Lead 2	SSR 1/120AC Live	Carries 120V	*Controlled by SSR 1 + Temperature controller 1 (chamber). Not directly plugged into 120AC live, needs a 24V signal to send power
4	Ext. Nozzle Hot Lead 2	SSR 2/120AC Live	Carries 120V	*Controlled by SSR 2 + Temperature controller 2 (ext nozzle). Not directly plugged into 120AC live, needs a 24V signal to send power
5	-	120AC Ground	Reduces shock hazard in event of short circuit	Directly plugged into 120AC ground
110 AC 				
*Currently 120AC is only used for the thermoplastics toolhead. If 120AC is needed for another toolhead application, a seperate 24V signal will be needed to be provided to the SSRs, since the temperature controllers will only provide the 24V signal when the "Thermoplastics Switch" is turned ON. <i>Be sure to make this alternate 24V signal fault tolerant, meaning it only works when the thermoplastics switch is OFF, else there is the possibility the SSRs send full power to the thermoplastics toolhead and cause damage.</i>				

5-Prong 24V DC Connector

This Cable is also Potentially Dangerous

This sheet is defining the black 5-pin cable carrying 24V of direct current (DC) going to the tool-head. **DO NOT CONFUSE IT FOR THE 120AC 5-PIN CABLE.** This cable can still carry significant amperage so it should be considered dangerous as well. The pins are defined 1-5 and can be seen in the diagram below and also on the end of the connectors themselves. The "connection" refers to the end of the cable that is wired to a device in the robot controller.

Connector Pin #	Toolhead Connection	Controller Connection	ABB Digital Output	Function (Thermoplastics)	Function (Ceramics)
1	-	-	24V Relay "RU1"	-	-
2	2 blower fans (+)	-	24V Relay "RU2"	Turns on 2 blower fans	-
3	2 other blower fans (+)	(CL57T) +Vdc	24V Relay "RU3"	Turns on other 2 blower fans	Turns on motor
4	Motor + chamber fans (+)	-	24V (+)	Motor + chamber cooling fans	-
5	All fans (-)	(CL57T) GND	24V Ground	Ground for all 24V components	Ground for all 24V components



12-Core I/O->ClearCore Cable

This sheet describes the gray signal cable located in the robot controller. *It is not the same as the 12-prong signal cable going to the tool head.*

This cable is used for carrying the ABB's I/O signals to the clearcore controller or other devices.

Wire Color	ABB Digital Output	Clearcore Input	Thermoplastics Function	Ceramics Function
Yellow	DO1	A-10	Start/Stop Extrusion	Start/Stop Extrusion
Purple	DO2	DI-6	-	-
Dark Brown	DO3	DI-7	-	-
Grey	DO4	DI-8	2 Blower Fans	-
Pink	DO5	A9	2 More Blower Fans	Turn on Motor
Blue	DO6	A-11	-	Reverse Motor Direction
White	AO1	A-12	Analog Motor Velocity/Extrusion	Analog Motor Velocity/Extrusion
Black	(unused)	-	-	-
Light Brown	(unused)	-	-	-
Orange	(unused)	-	-	-
Green	(unused)	-	-	-
Red	(unused)	-	-	-

ABB I/Os							
This sheet is for describing the path and function of the ABB I/Os for easy diagnosing should the system malfunction.							
For example, the digital ABB output "DO7" travels to the Teknic ClearCore controller input "A-9". When this signal is triggered, it activate's the ClearCore's output pin "I/O-2". This in turn activates a 24V relay "X3", which allows 24V to pass through the 24V-DC 5-prong cable.							
If the thermoplastics tool-head is hooked up, this would turn on 2 of the 4 blower fans. If the cermaic extruder is hooked up, this would turn the motor on.							
Signal Name	ClearCore Input	ClearCore Output	24V Relay	24V 5-prong	Function (Thermoplastics)	Function (Ceramics)	
DO1	A-10	M-0, I/O-0	-	-	Start/Stop Extrusion	Start/Stop Extrusion	
DO2	DI-6	I/O-5	X0	-	-	-	
DO3	DI-7	I/O-4	X1	#1	-	-	
DO4	DI-8	I/O-3	X2	#2	Fans_1	-	
DO5	A-9	I/O-2	X3	#3	Fans_2	Turn on Motor	
DO6	A-11	I/O-1	-	-	-	Reverse Motor Direction	
DO7	-	-	-	-	-	-	
DO8	-	-	-	-	-	-	
AO1	A-12	M-0, I/O-0	-	-	Variable Motor Velocity	Variable Motor Velocity	
AO2	-	-	-	-	-	-	
AO3	-	-	-	-	-	-	
AO4	-	-	-	-	-	-	
These inputs (DI-6/7/8/A-9) can instead be directly wired directly to the 24V							
relay (X0/1/2/3) if additional Clearcore pins are needed							